In-Class Challenge: Iterating Through Data Structures

Course: Data Science and Data Visualization
Topic: Data Wrangling — Iterating Through Data Structures
Dataset: UCI_Online_Retail_7500row_practice.csv

Scenario

You are a junior data analyst working for an online retail company. Your manager wants you to investigate the retail transaction data and prepare a small data-quality and sales summary.

Your goal: Use **lists, dictionaries, loops, and pandas DataFrames** to explore and process the dataset.

Challenge 1 — Explore the Dataset

Load the CSV file into a pandas DataFrame.

Answer:

1. How many rows and columns are in the dataset?
2. What are the column names?
3. Which columns contain numerical data?
4. Which columns contain categorical/text data?

Hint: Try .shape, .columns, .head(), and .info().

Challenge 2 — Work with a List

Create a Python **list** containing all column names.

Then use a for loop to display each column name with its position.

Hint: Use list(df.columns) and enumerate().

Challenge 3 — Investigate Missing Data

Determine how many missing values exist in each column.

Then create a **dictionary** where:

- Key → column name
- Value → number of missing values

Hint: Start with an empty dictionary {} and iterate through df.columns.

Challenge 4 — Create a Product Lookup Dictionary

Create a dictionary that connects:

StockCode → Description

For example, the dictionary should allow you to look up the description of a product when you know its stock code.

Hint: First investigate: df[["StockCode", "Description"]]

Challenge 5 — Calculate Sales Value

Create a new column called: **SalesValue**

Calculate it using: Quantity × UnitPrice

Hint: Try the pandas column operations you learned in class.

Challenge 6 — Practice Iteration

Create the same SalesValue calculation using a **loop**.

Store the calculated values in a Python list and then add the list as a new DataFrame column.

Hint: You can iterate through DataFrame rows using:

for index, row in df.iterrows():

Final Challenge — Retail Analyst

The manager wants to identify the company's best-performing products.

Before analyzing the products:

1. Exclude cancelled transactions.
2. Remove records where CustomerID is missing.
3. Calculate SalesValue.
4. Create a dictionary containing: **StockCode → Total SalesValue**
5. Find the **top 5 products by total sales value**.

Hints

- Cancelled invoices can be identified from InvoiceNo.
- Use a new DataFrame for your cleaned data.
- Start your product-sales dictionary as {}.
- Use an if statement to determine whether a product is already in the dictionary.
- Python's max() function may help you identify the largest value.
- For the top 5, think about how you can repeatedly identify the largest values or convert your dictionary into another useful data structure.